STAGE 2 — COMPREHENSIVE

ember_ind

Order 2 | metric / inter-timestep / aggregates

```
T (records)         : 26
D (carriers)        : 8
Carriers            : Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Solar, Wind
Engine              : cnt 2.0.4
Schema              : 2.1.0
Source SHA-256      : 05cde94fa8f37b98ae695073605eb97b...
Content SHA-256     : ad0da5d361bc14138f450b838ba2783e...
```

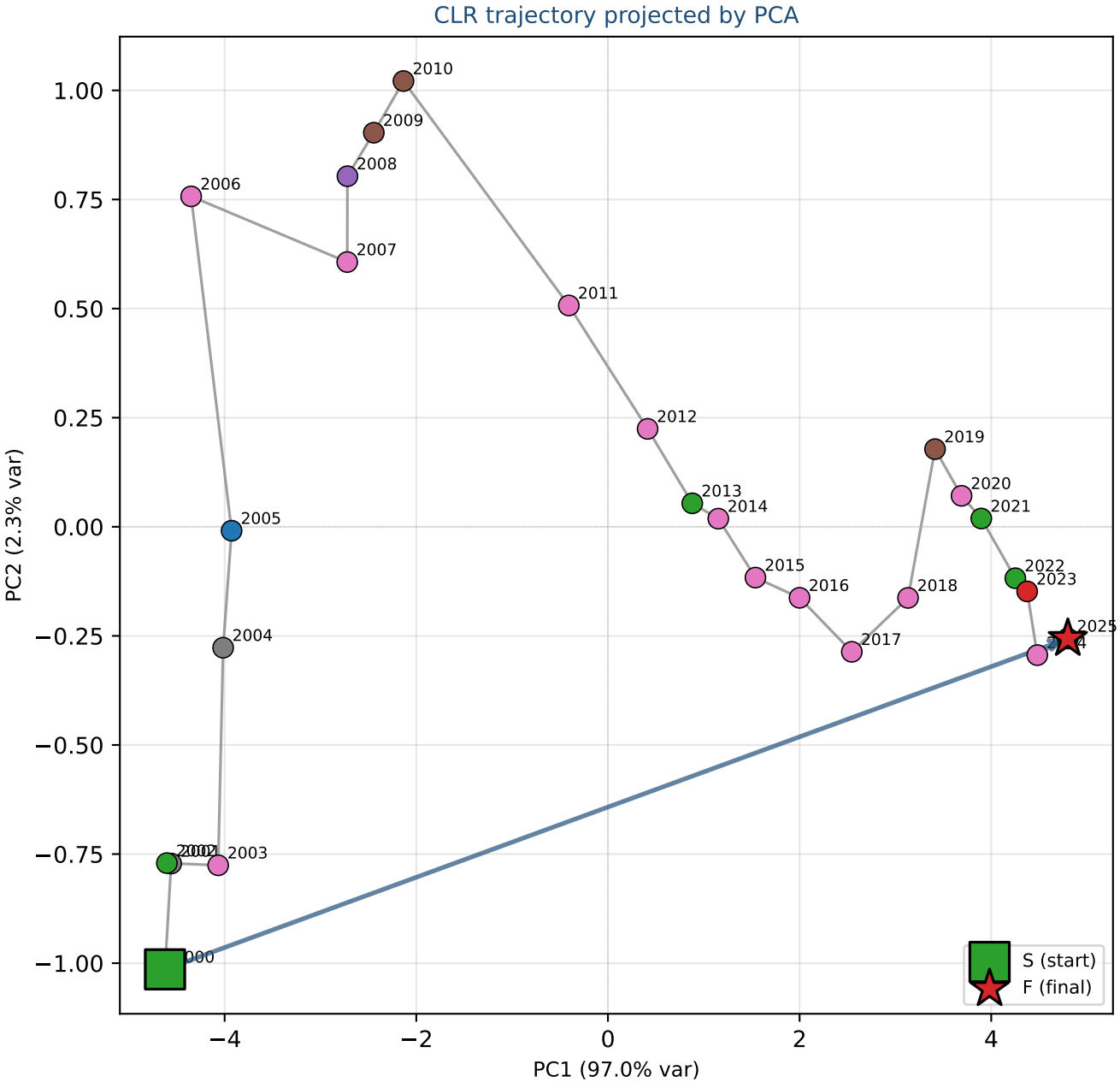
Doctrine: integer orders only. Order 1 = single composition.
Order 2 = anything aggregating across multiple timesteps.

This Stage 2 atlas covers:

- Trajectory aggregates (Hs, Aitchison-norm, ILR-axes)
- Inter-timestep metrics (ω, κ, helmsman, ring, locks)
- Pairwise structure (variation, correlation, divergence)
- System Course Plot (CNT Math Handbook Ch 15)
- Compositional Bearing Scope CBS (Ch 16)
- Numeric summary

STAGE 2 | Plate 2 | System Course Plot — PCA 2D path with S→F net vector

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

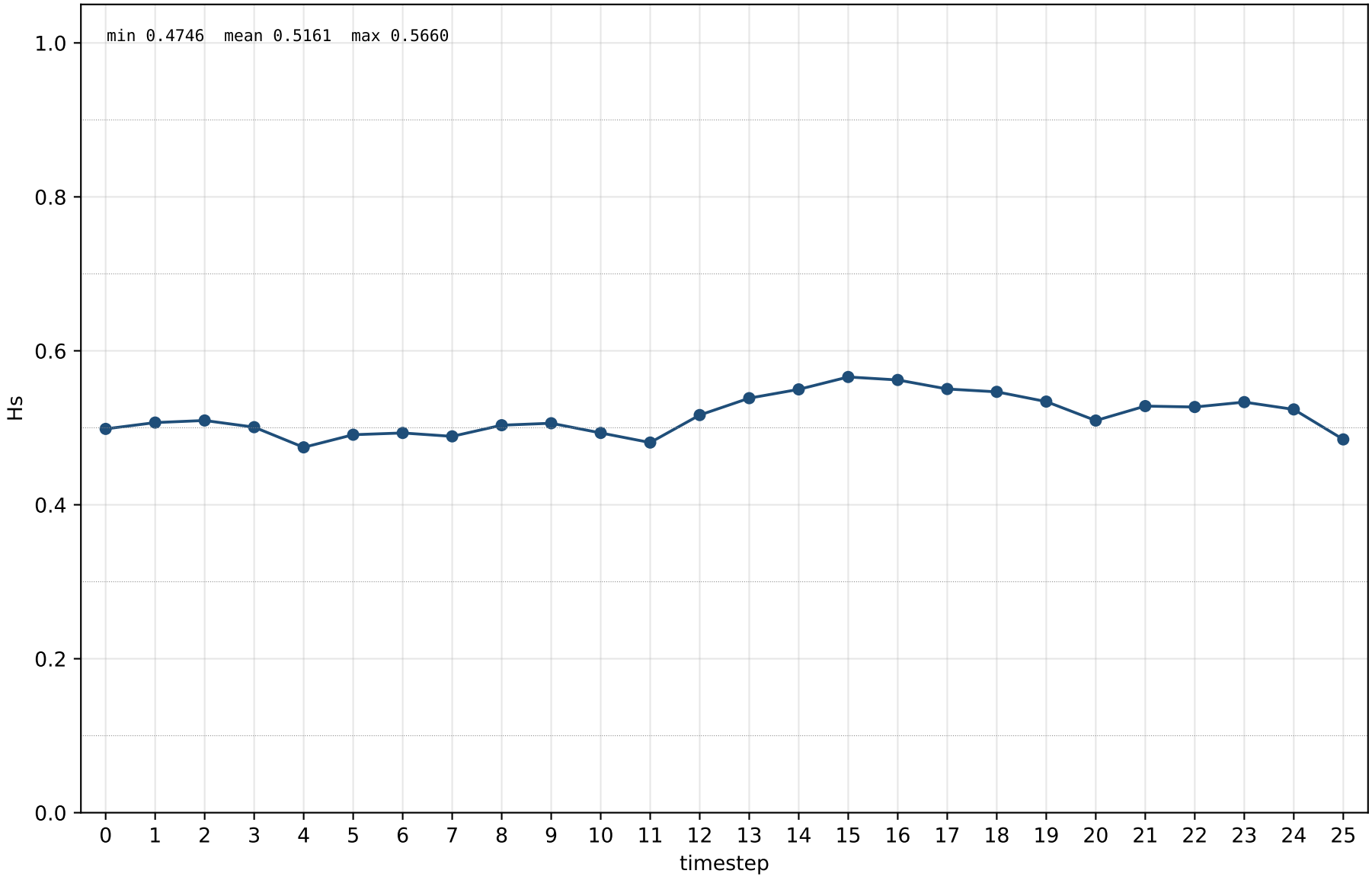


Navigation metrics (V_net = h_F − h_S)

```
h_start label    : 2000

h_final label    : 2025


net_distance     : 9.4603 HLR

path_length      : 14.6188 HLR

course_directness: 0.6471

  (1.0 = straight line, 0 = pure looping)


dynamic range S  : 10.5719 HLR

dynamic range F  : 5.8848 HLR


Reading guide:

  tight cluster  → stable composition

  long excursion → structural shift

  sharp turn     → regime transition

  loop           → cyclic behaviour

  spiral         → systematic drift
```

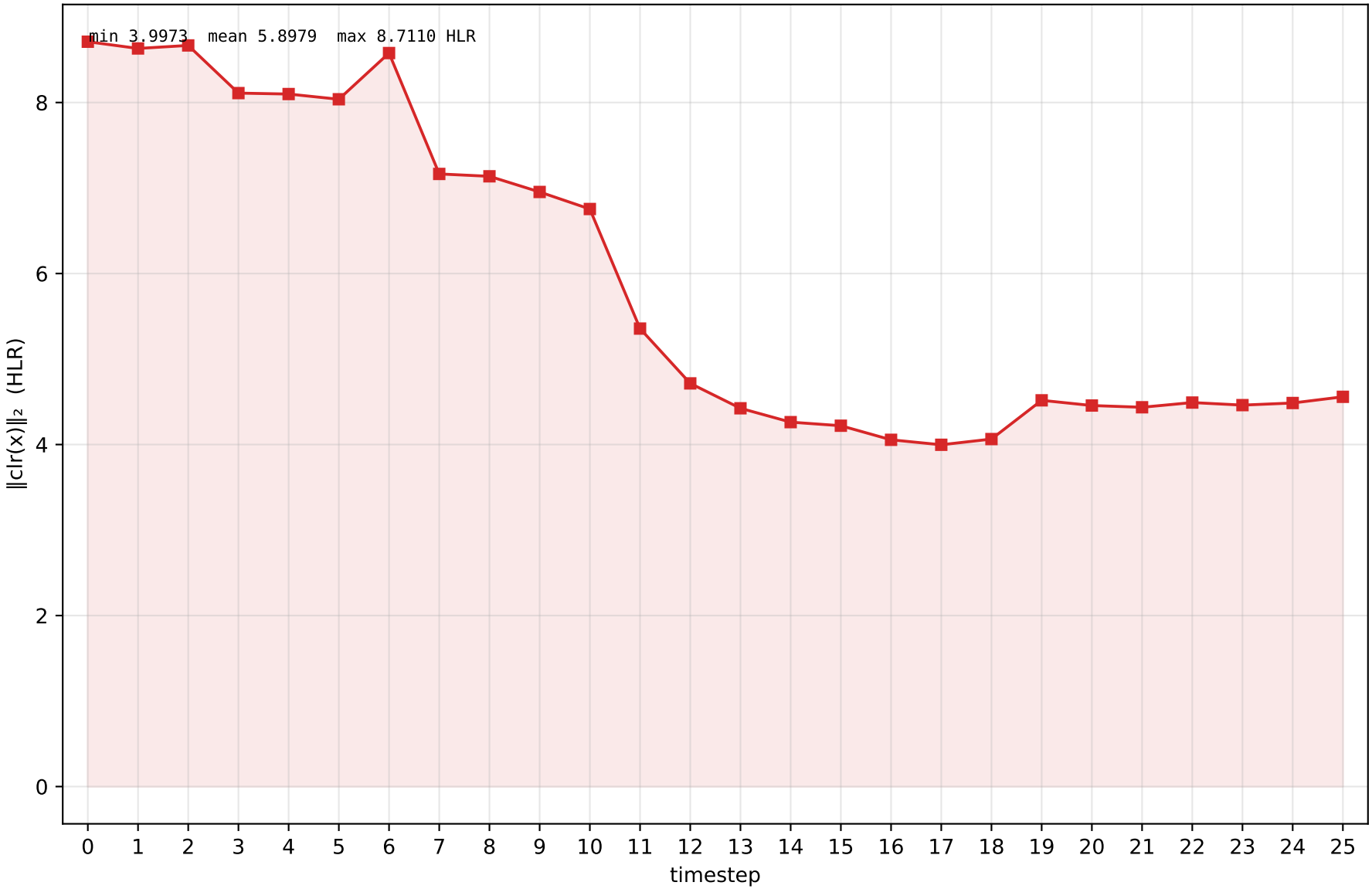
STAGE 2 | Plate 3 | Hs (Higgins scale) trajectory — Order 2 aggregate

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



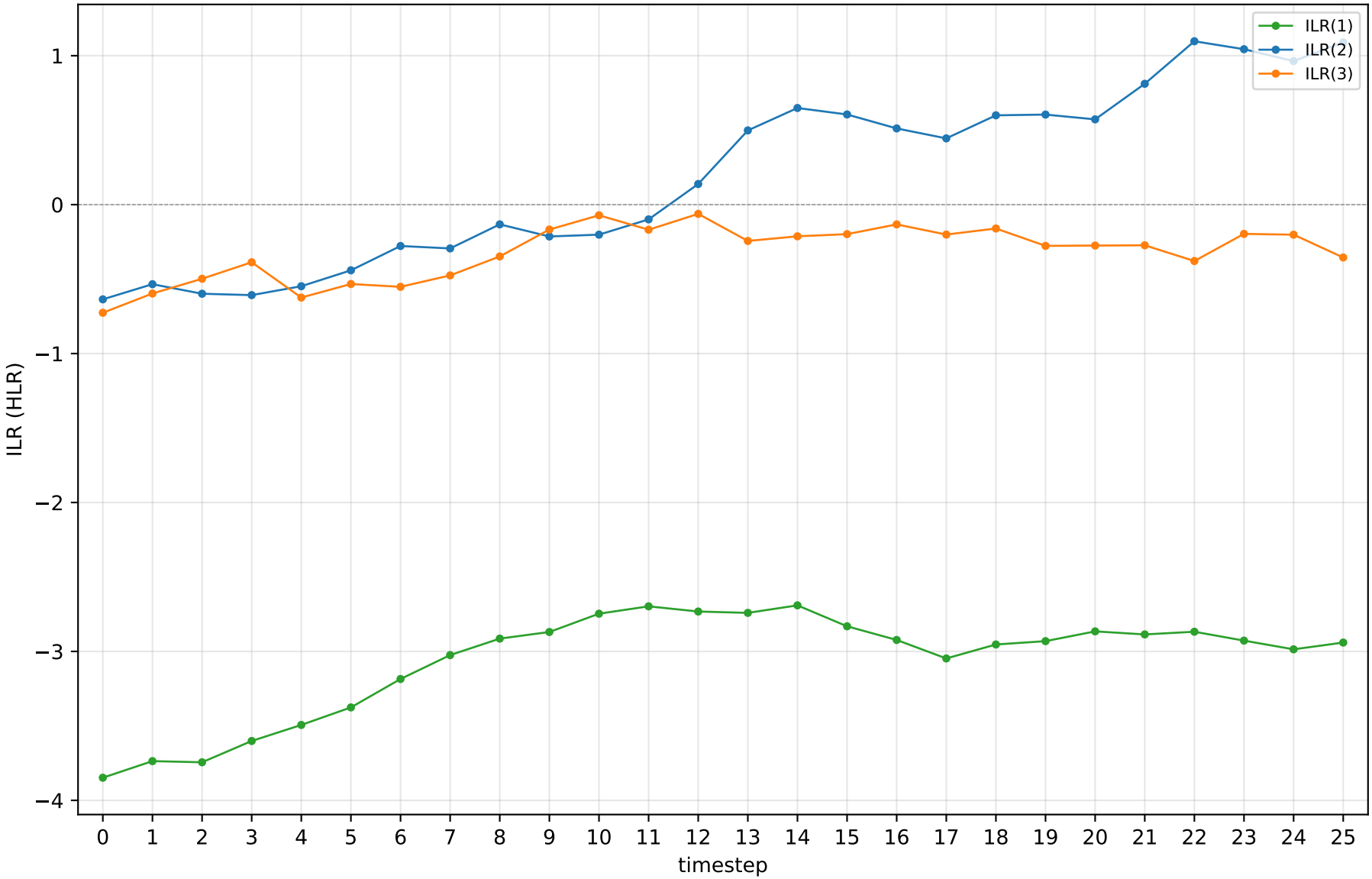
STAGE 2 | Plate 4 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



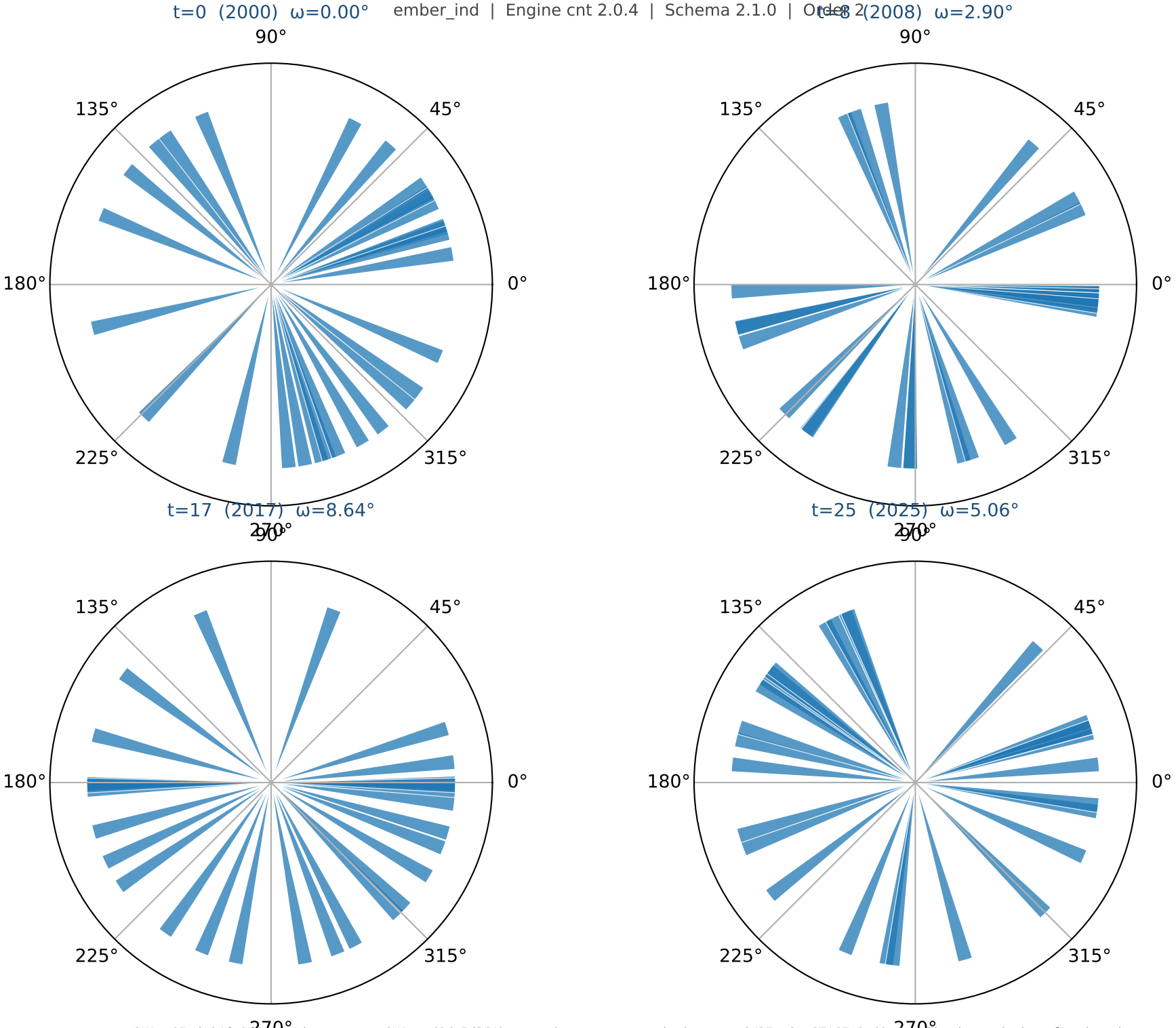
STAGE 2 | Plate 5 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



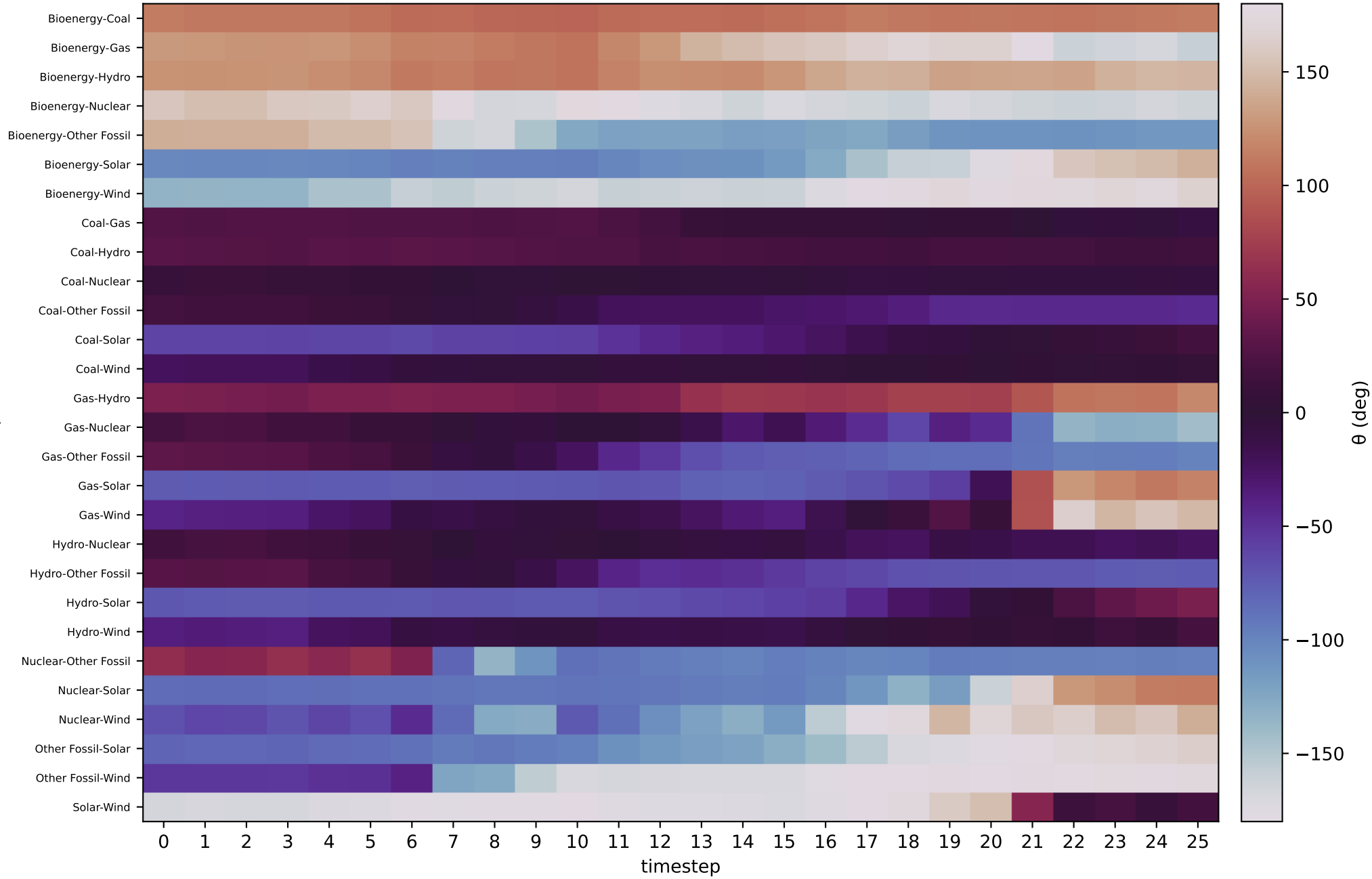
STAGE 2 | Plate 6 | Polar bearing rose (4 sample timesteps)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



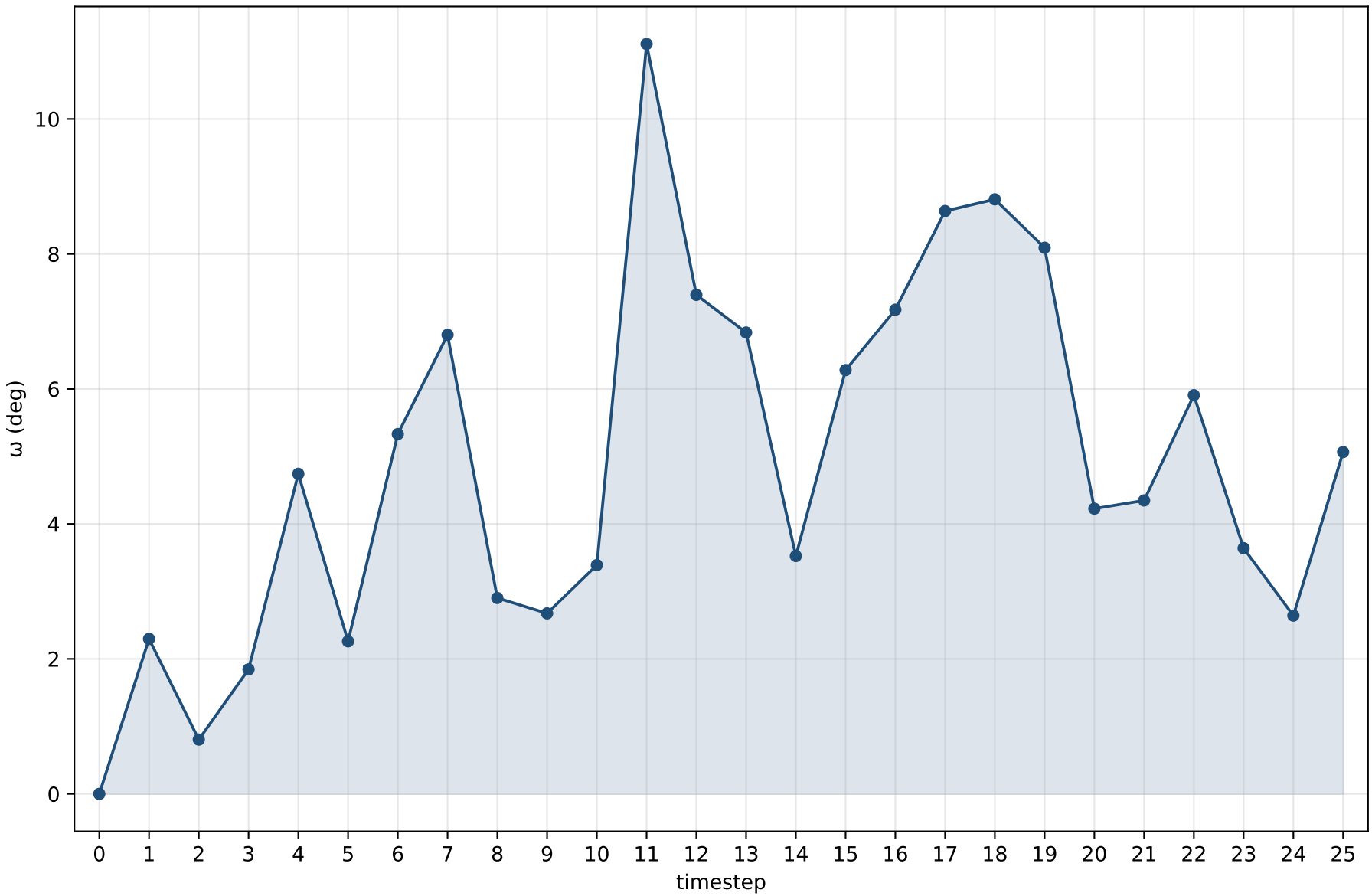
STAGE 2 | Plate 7 | Bearing heatmap T × C(D,2) pairs

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



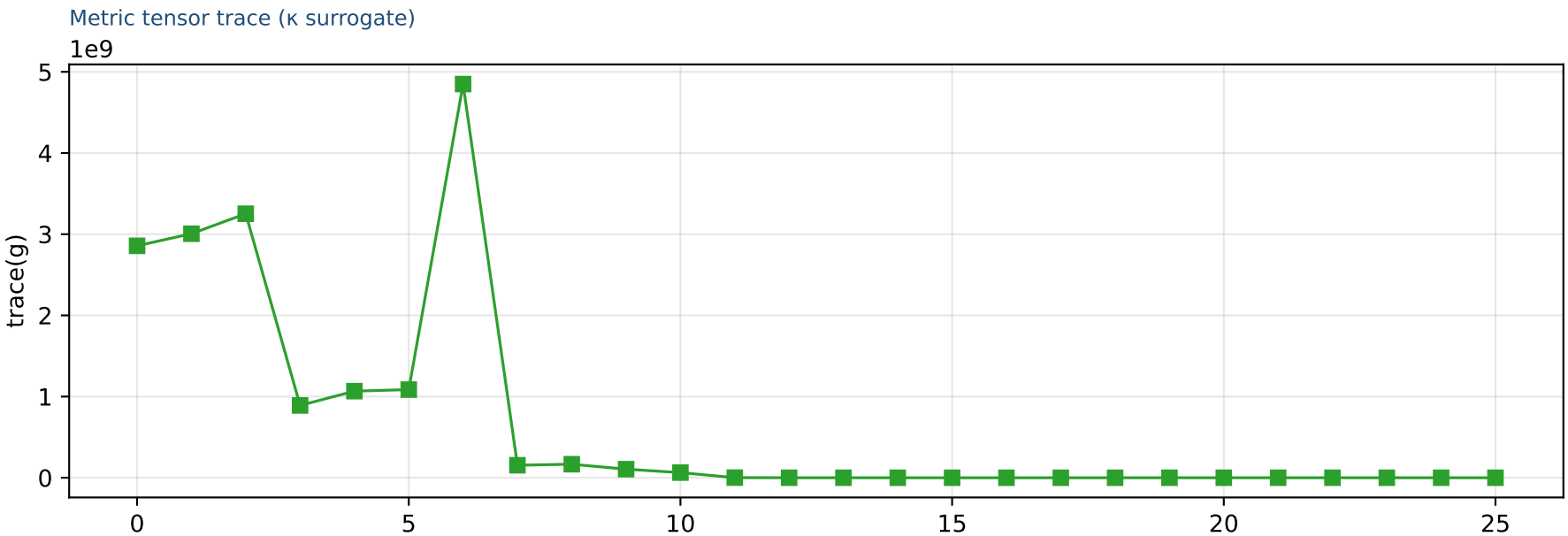
STAGE 2 | Plate 8 | ω angular velocity trajectory

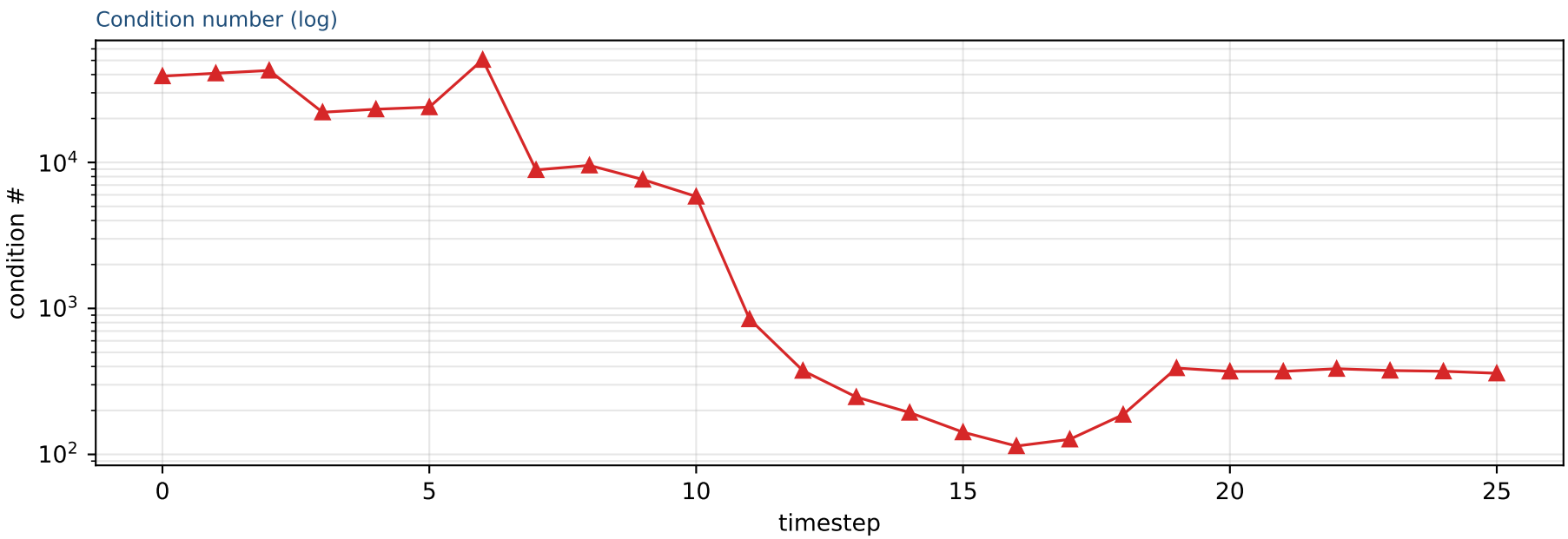
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 9 | Metric tensor trace + condition number

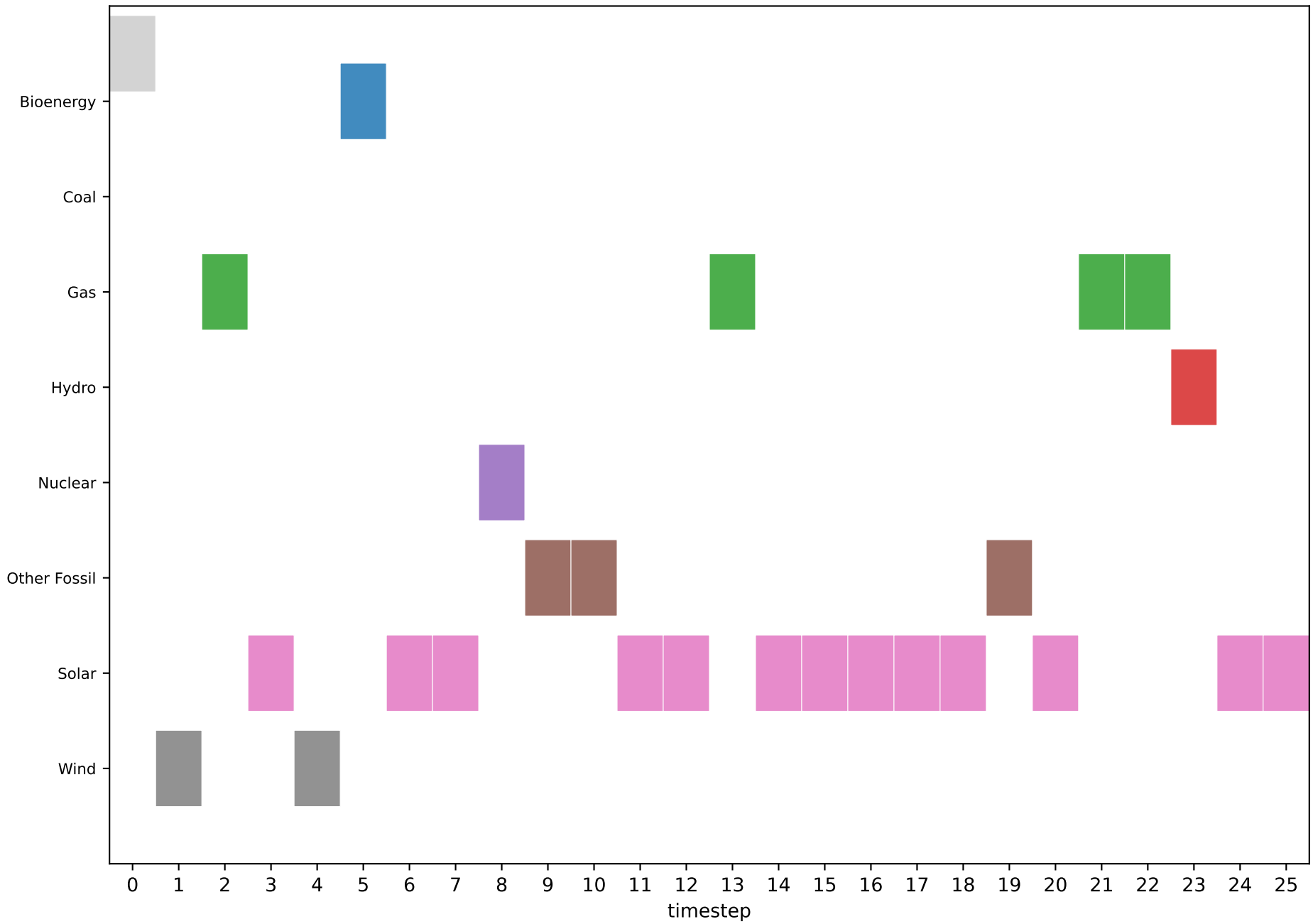
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2





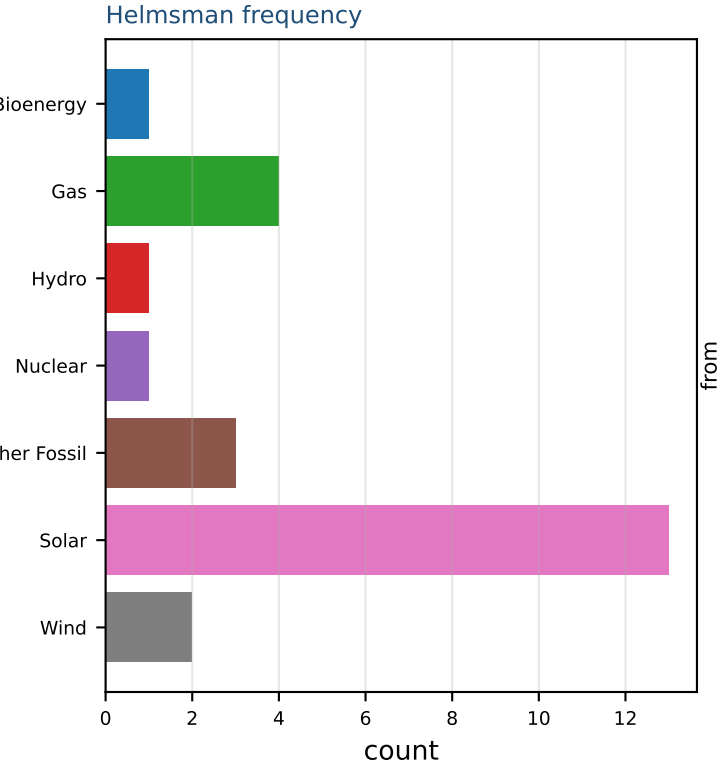
STAGE 2 | Plate 10 | Helmsman σ lineage timeline

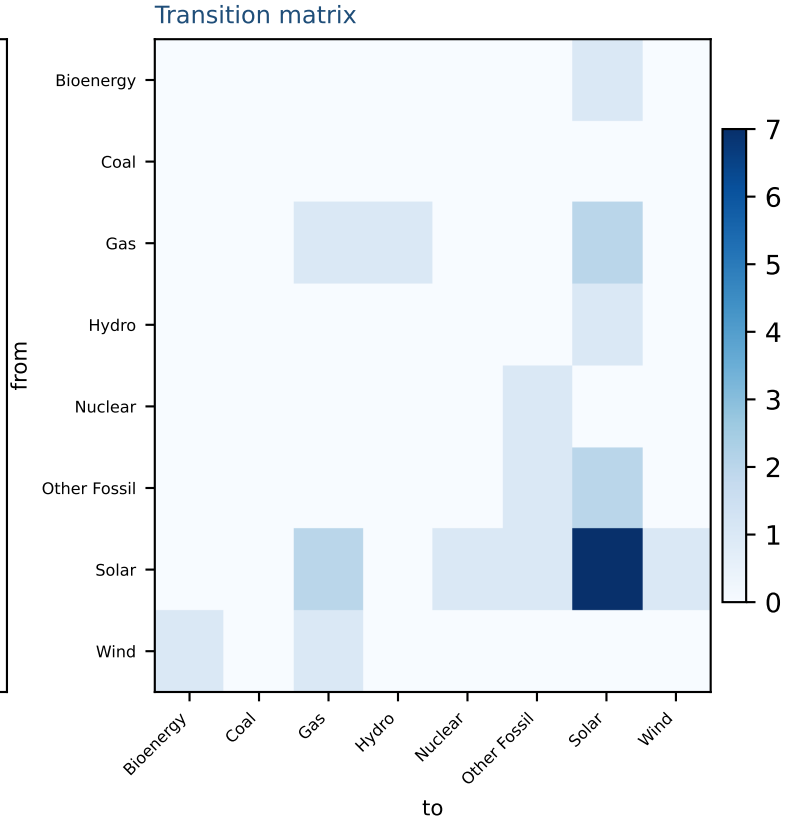
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

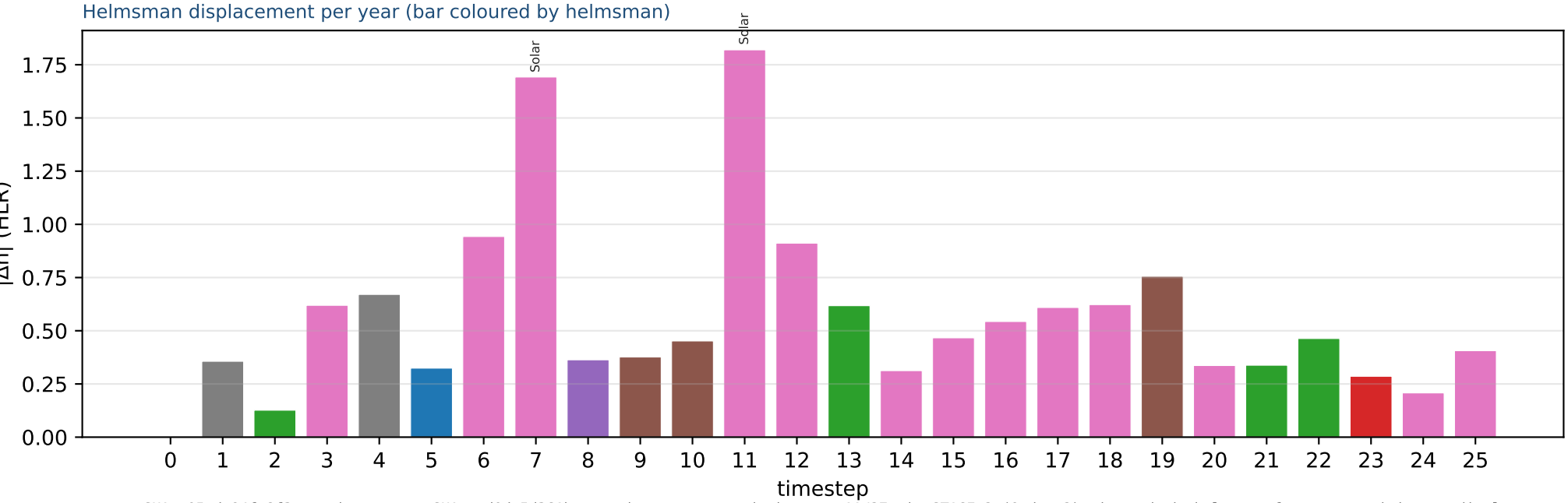


STAGE 2 | Plate 11 | Helmsman σ — frequency · transitions · per-year displacement

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

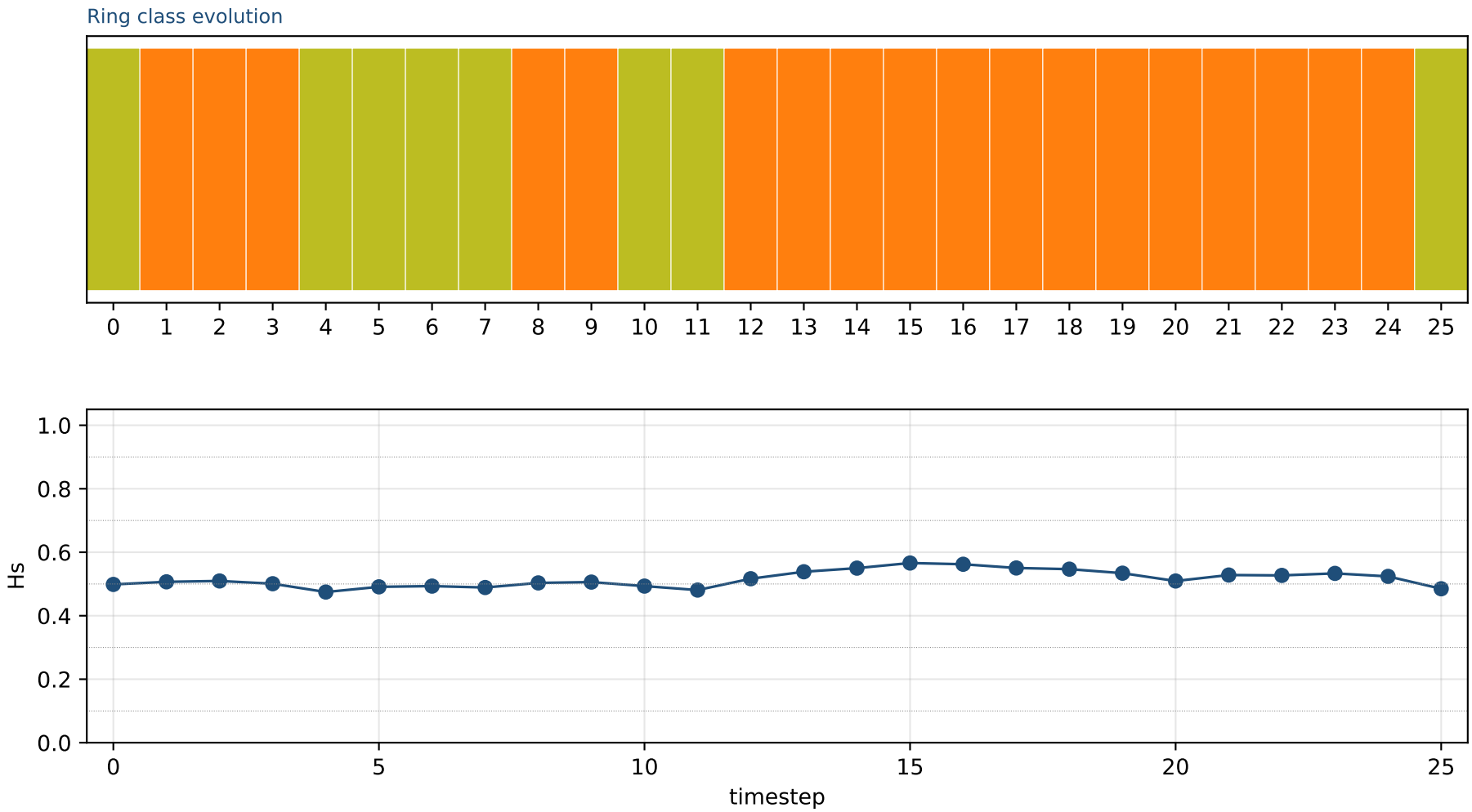






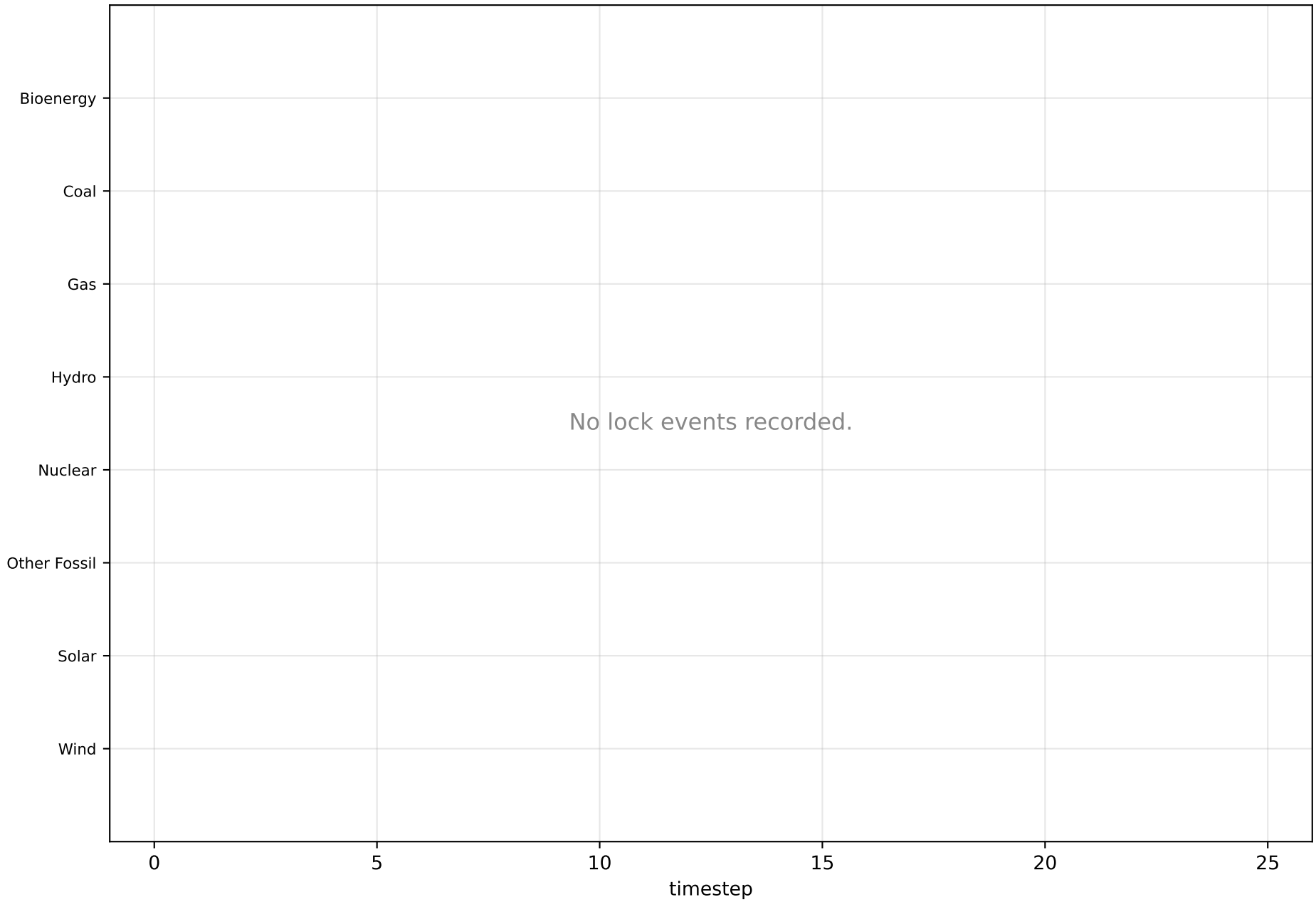
STAGE 2 | Plate 12 | Ring class evolution + Hs

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



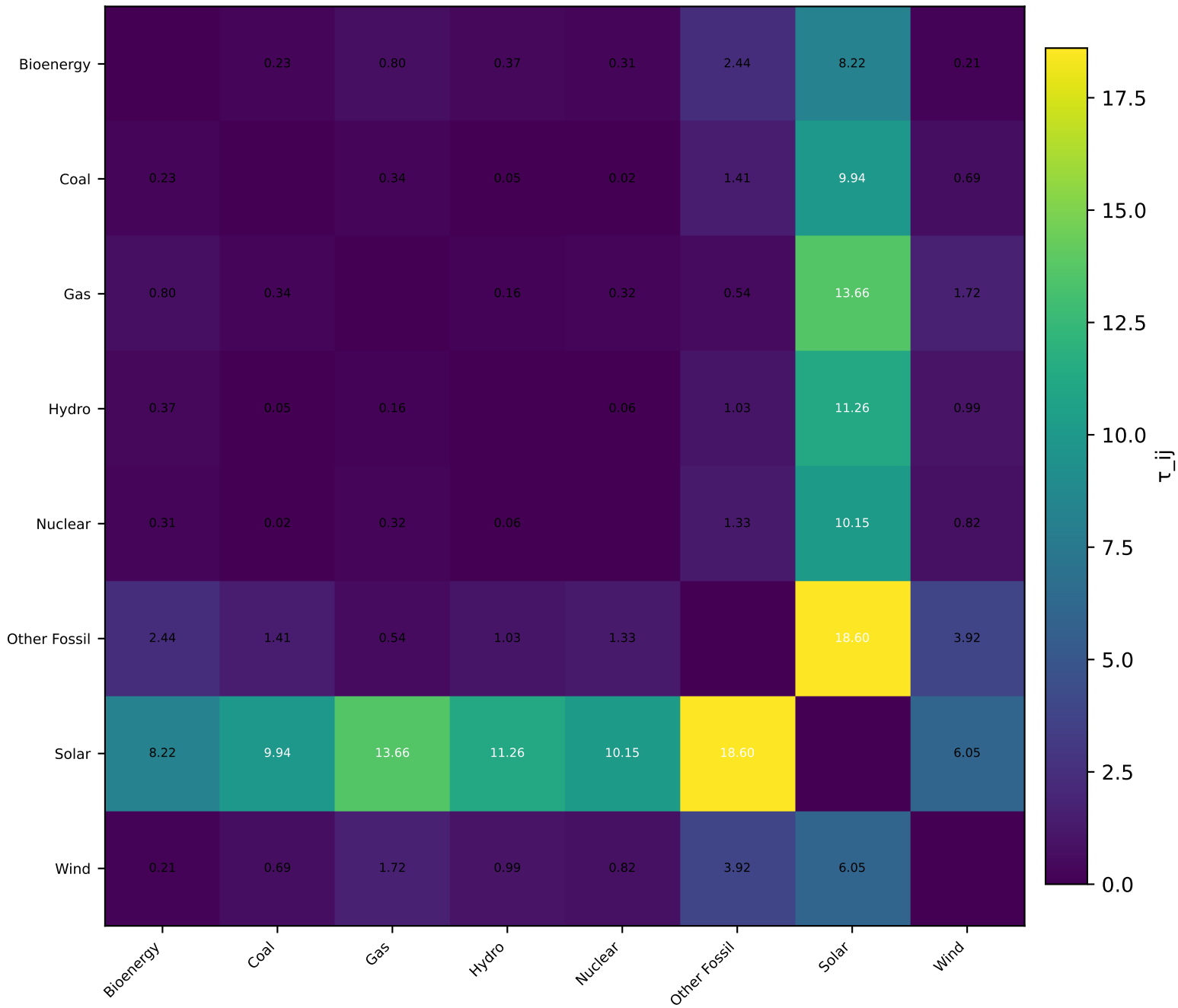
STAGE 2 | Plate 13 | Lock event timeline (carriers × timesteps)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



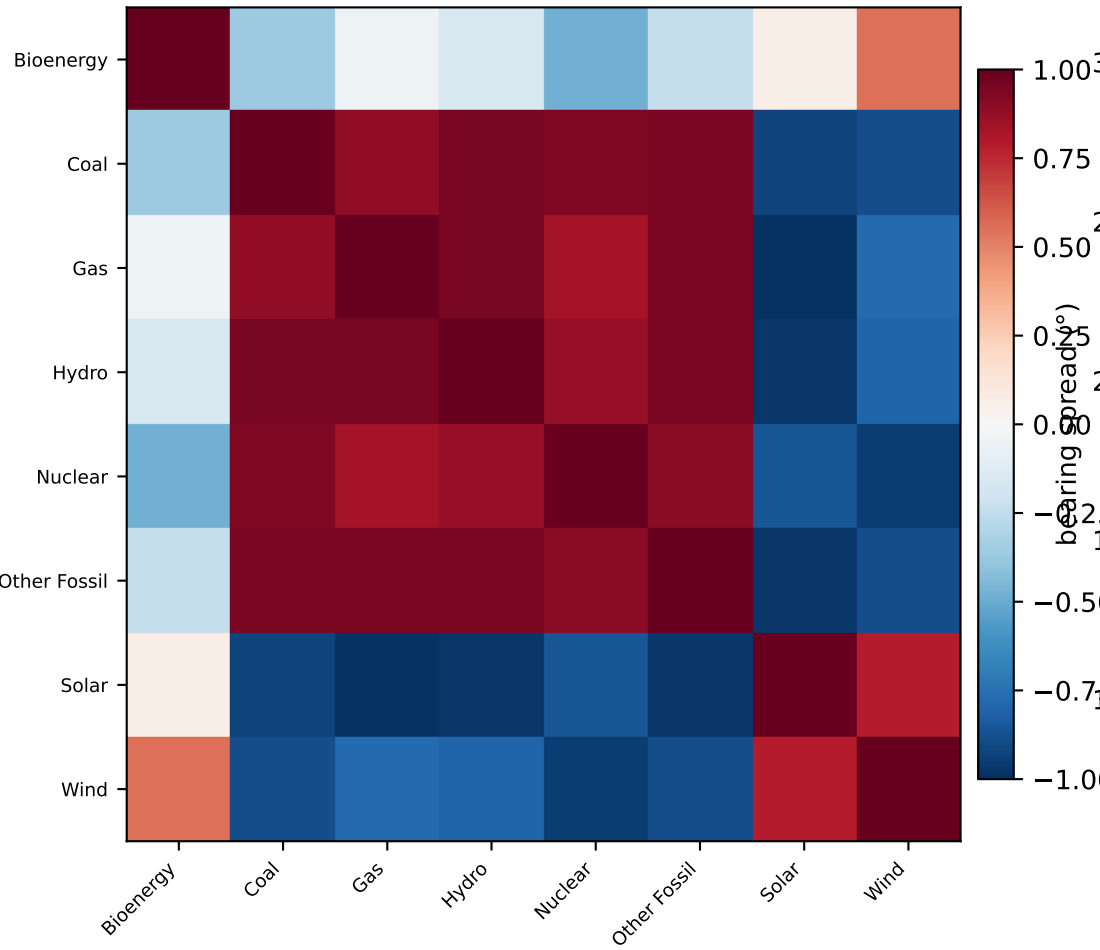
STAGE 2 | Plate 14 | Variation matrix τ_ij heatmap

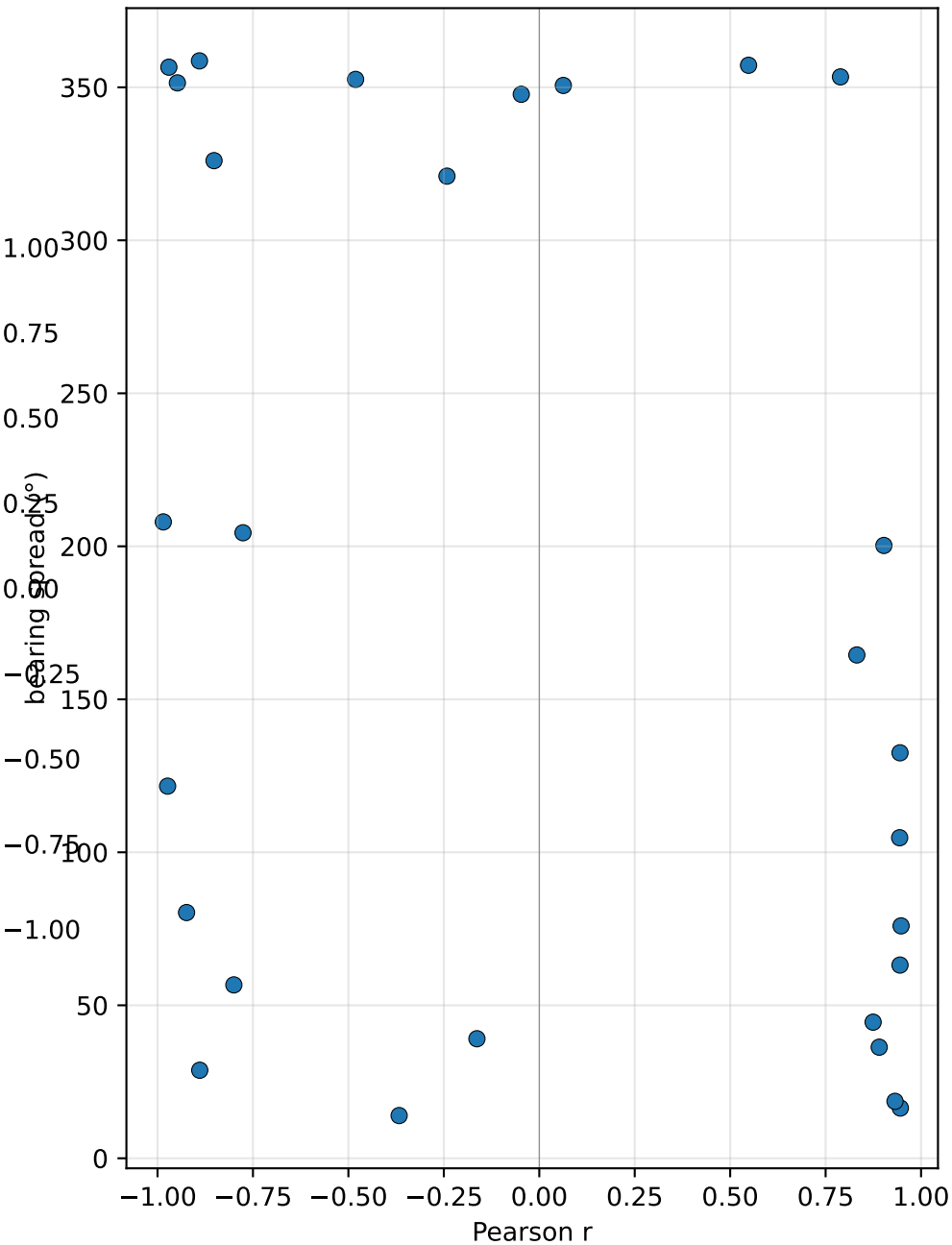
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 15 | Pair Pearson r heatmap + co-movement scatter

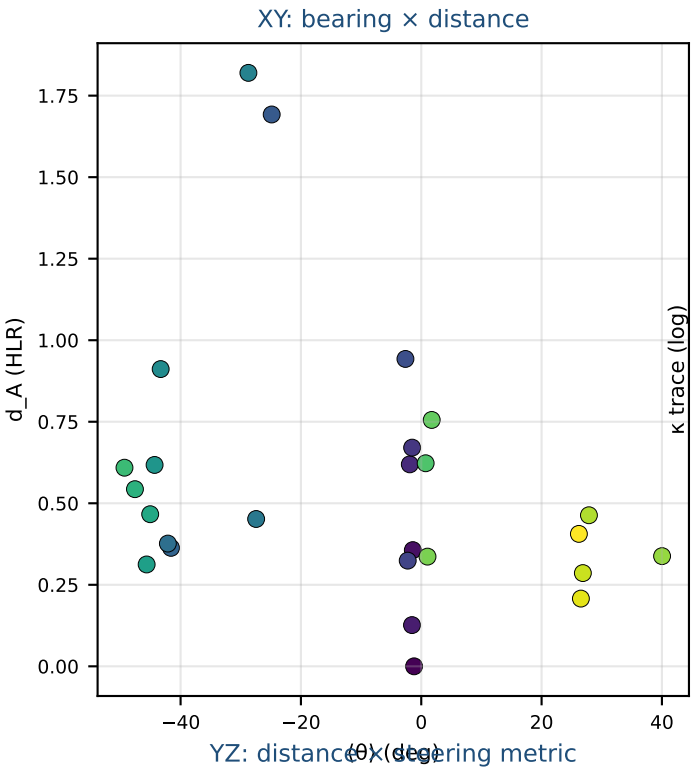
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

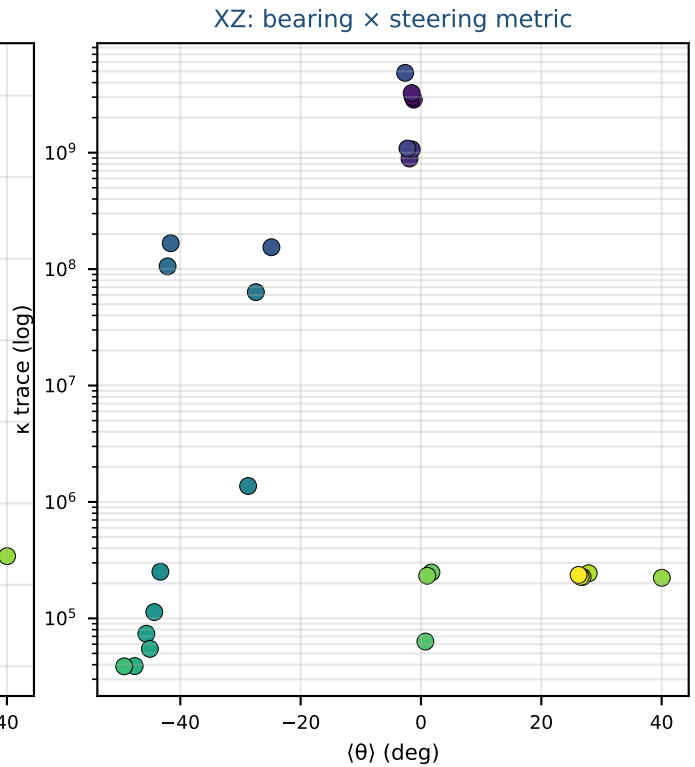


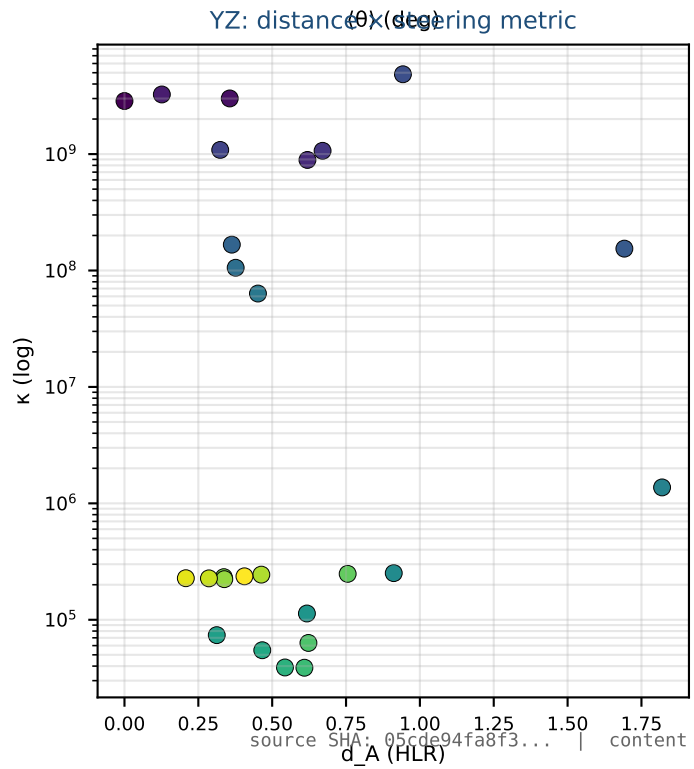


STAGE 2 | Plate 16 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2





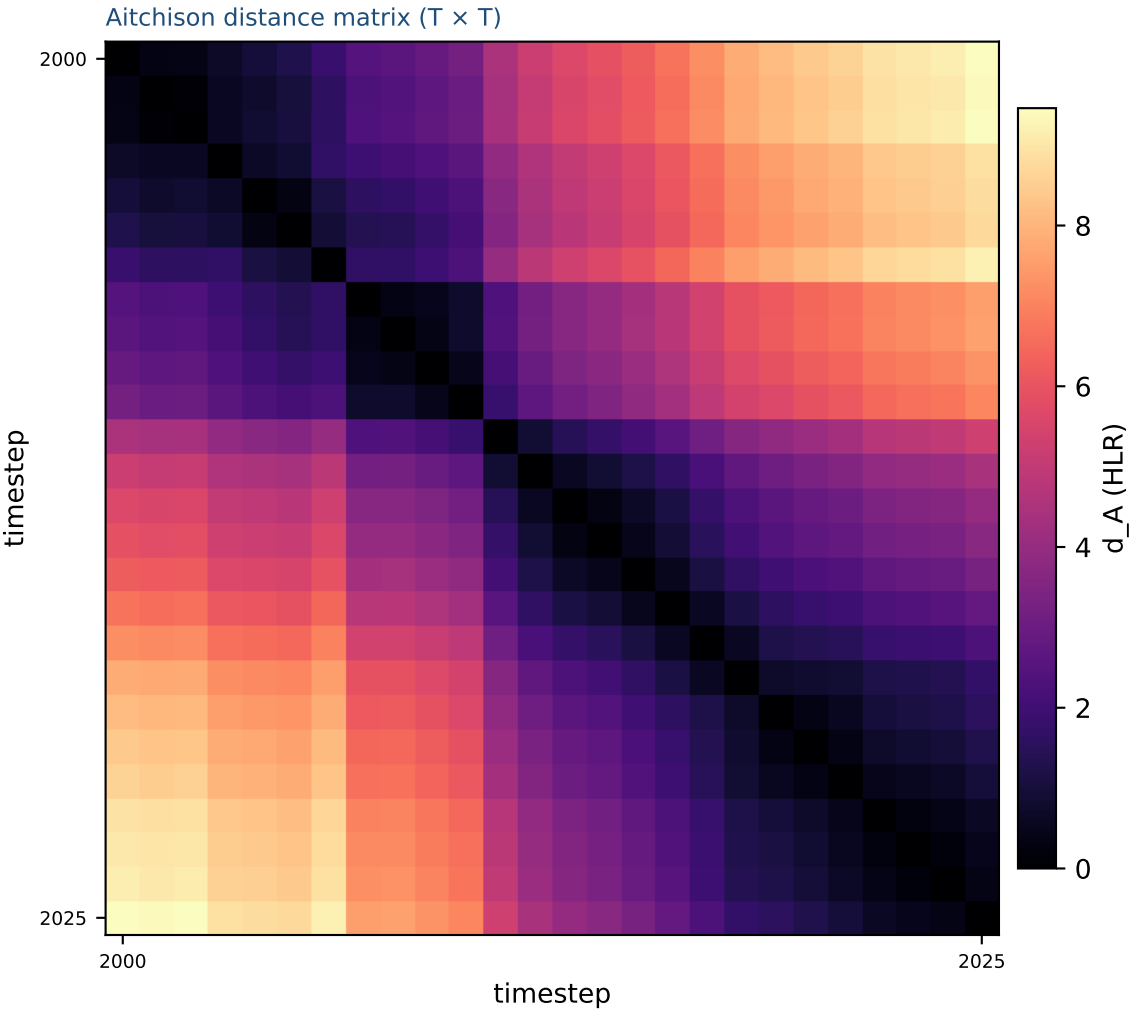


CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [-49.3, 40.0]°
d_A range  : [0.0000, 1.8198] HLR
κ range    : [3.87e+04, 4.85e+09]
Hs range   : [0.4746, 0.5660]
ω max      : 11.11°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

STAGE 2 | Plate 17 | Pairwise divergence — distance matrix + top-15 ranking

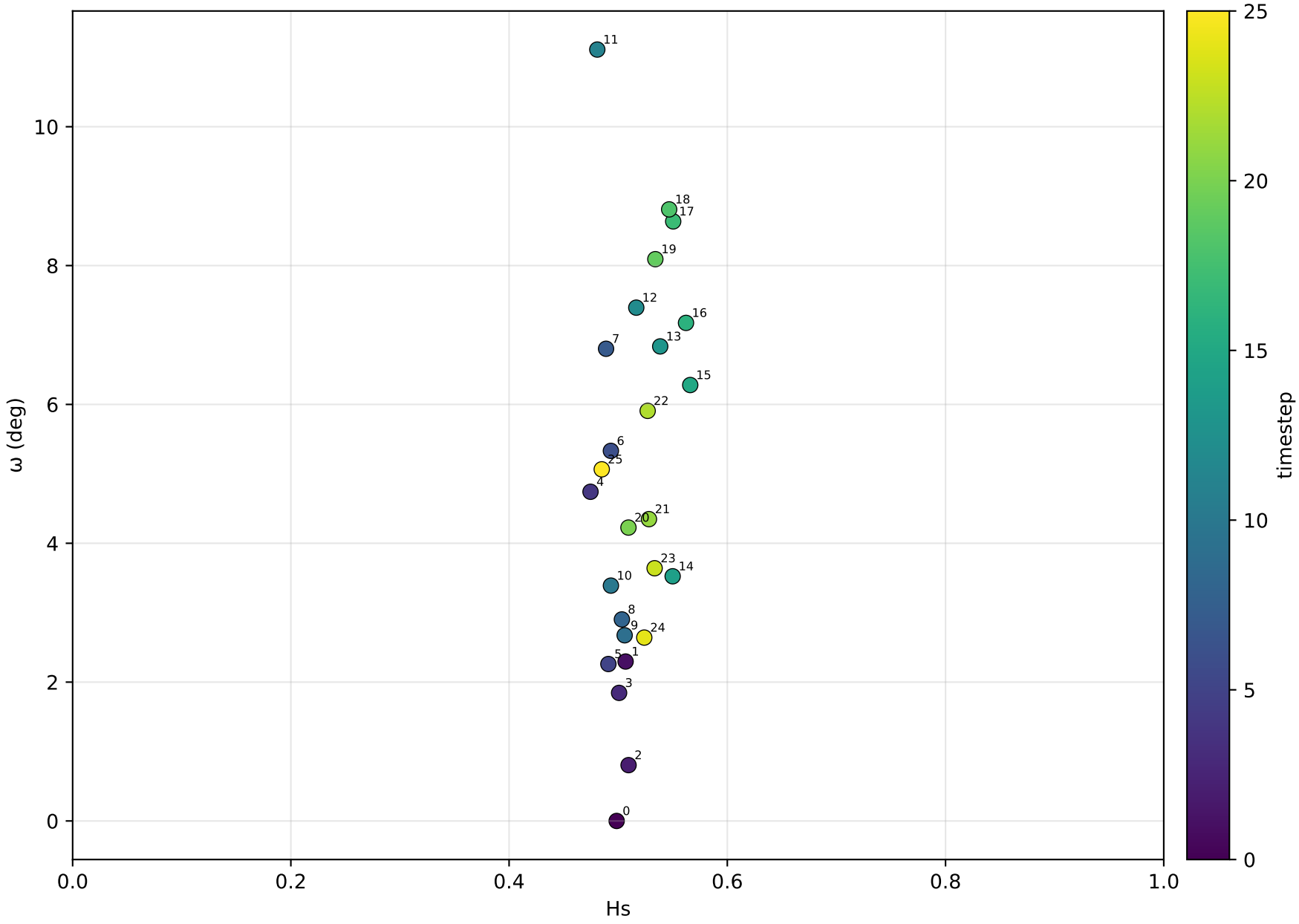
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



rank	A	B	d_A
1	2000	2025	9.4603
2	2002	2025	9.4248
3	2001	2025	9.3775
4	2006	2025	9.2075
5	2000	2024	9.1358
6	2002	2024	9.0976
7	2001	2024	9.0528
8	2000	2023	9.0441
9	2002	2023	9.0023
10	2001	2023	8.9569
11	2000	2022	8.9232
12	2006	2024	8.8998
13	2003	2025	8.8960
14	2002	2022	8.8848
15	2001	2022	8.8372

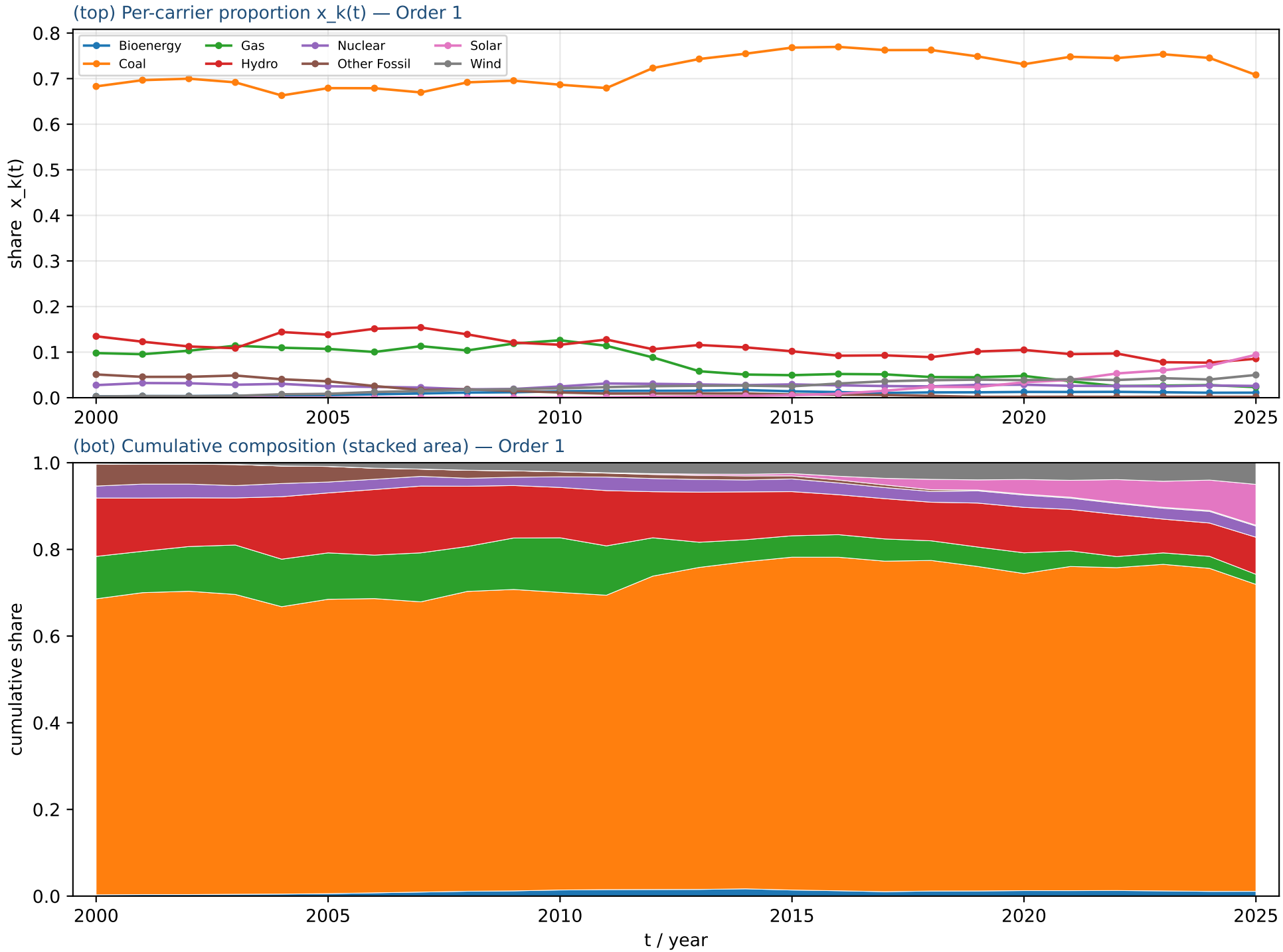
STAGE 2 | Plate 18 | ω vs Hs scatter (XY test)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 19 | Evolution of Proportions (raw + cumulative)

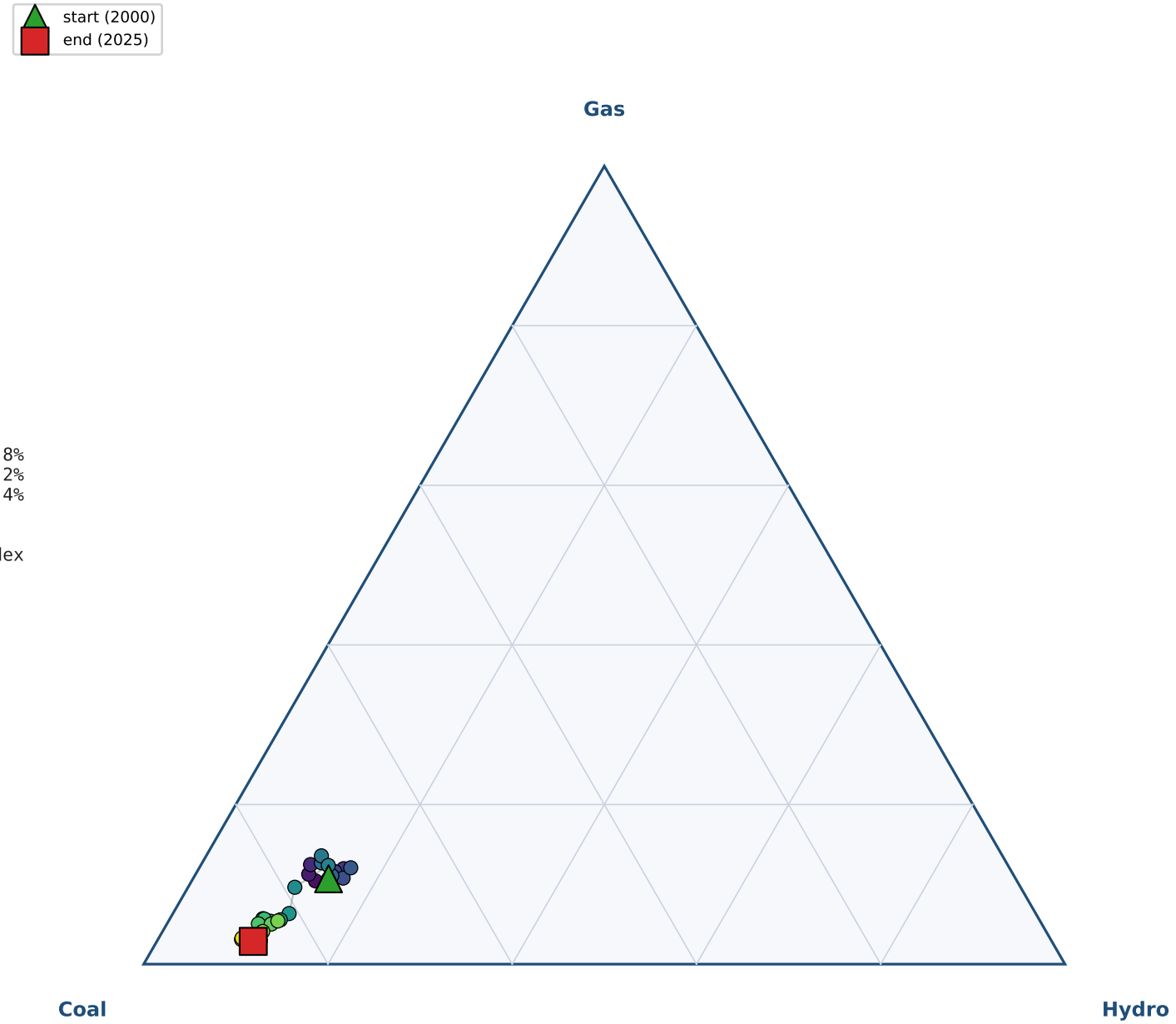
ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



STAGE 2 | Plate 20 | Ternary Diagram (top-3 carriers)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

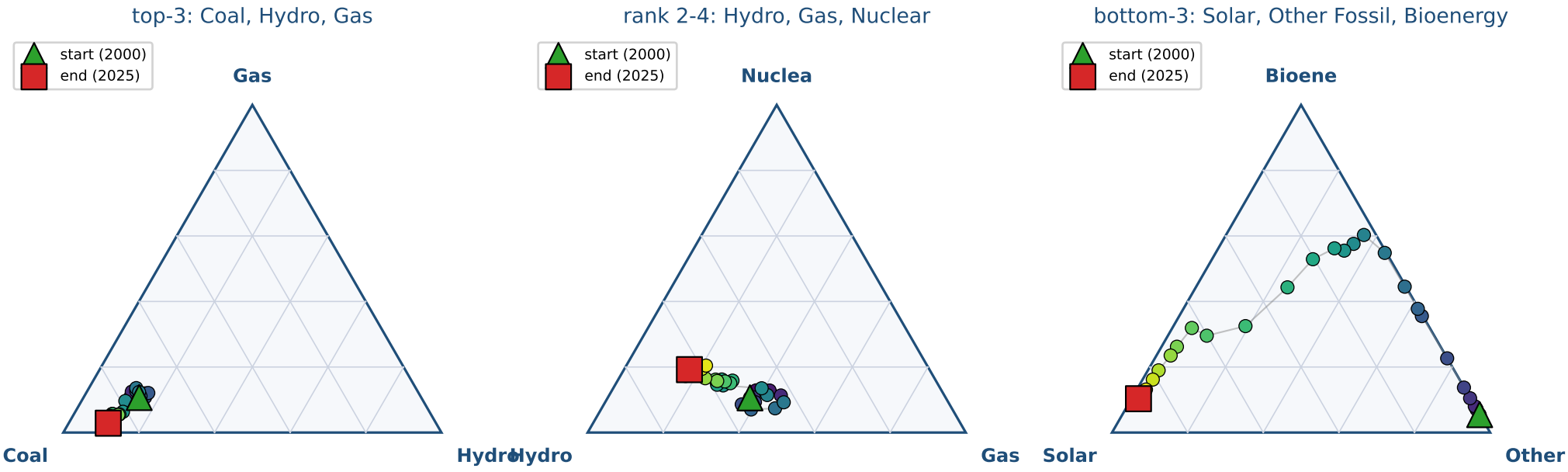
3-part subcomposition closed over Coal, Hydro, Gas
(remaining 5 carriers absorbed into closure)




```
Top-3 by mean share:
  Coal            71.8%
  Hydro           11.2%
  Gas              7.4%

T = 26 timesteps
colour map = time index
▲ start  ■ end
```

STAGE 2 | Plate 21 | Three-Part Subcompositions (3 ternaries)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

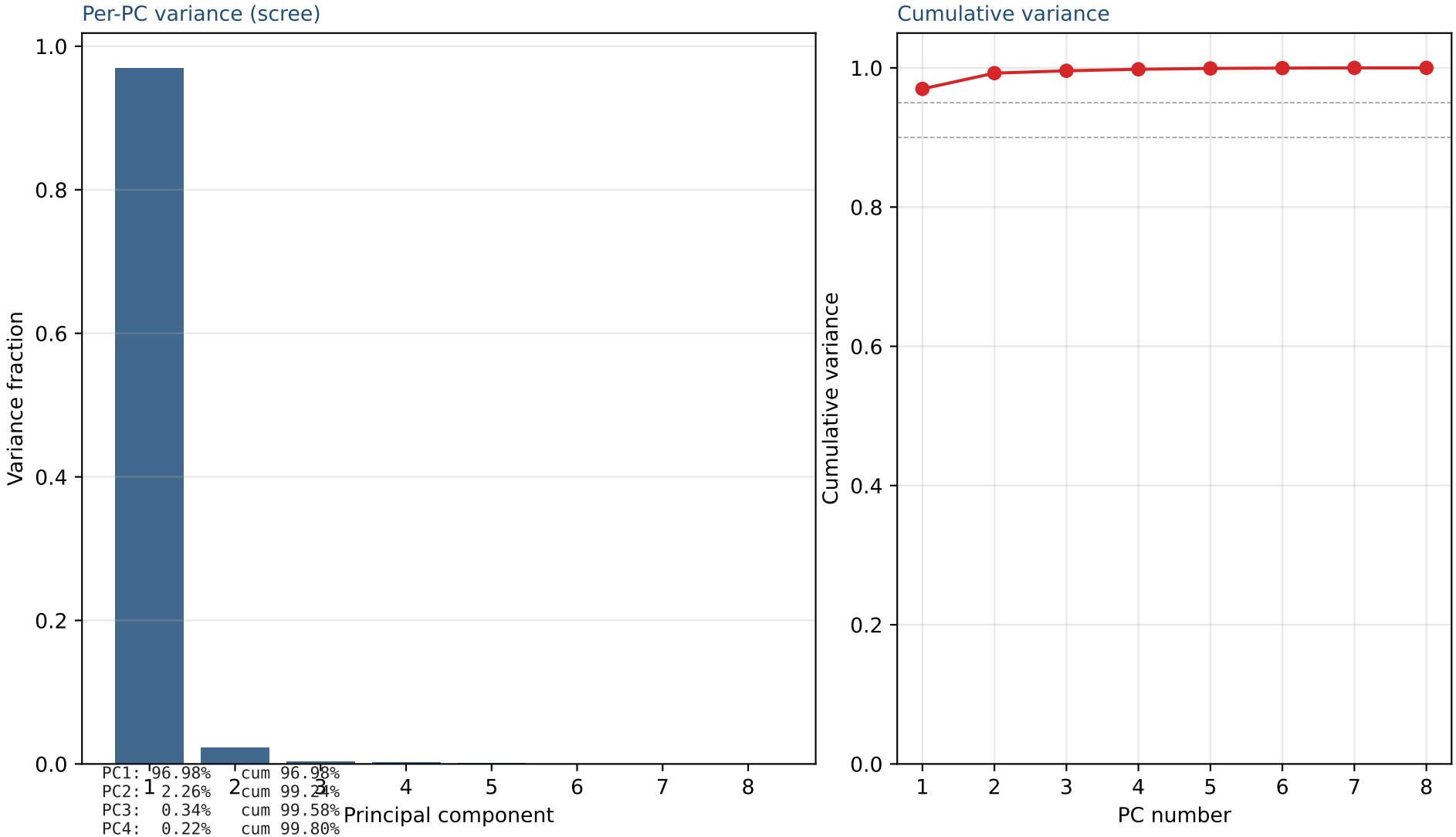


Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

STAGE 2 | Plate 22 | Scree Plot — CLR PCA

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



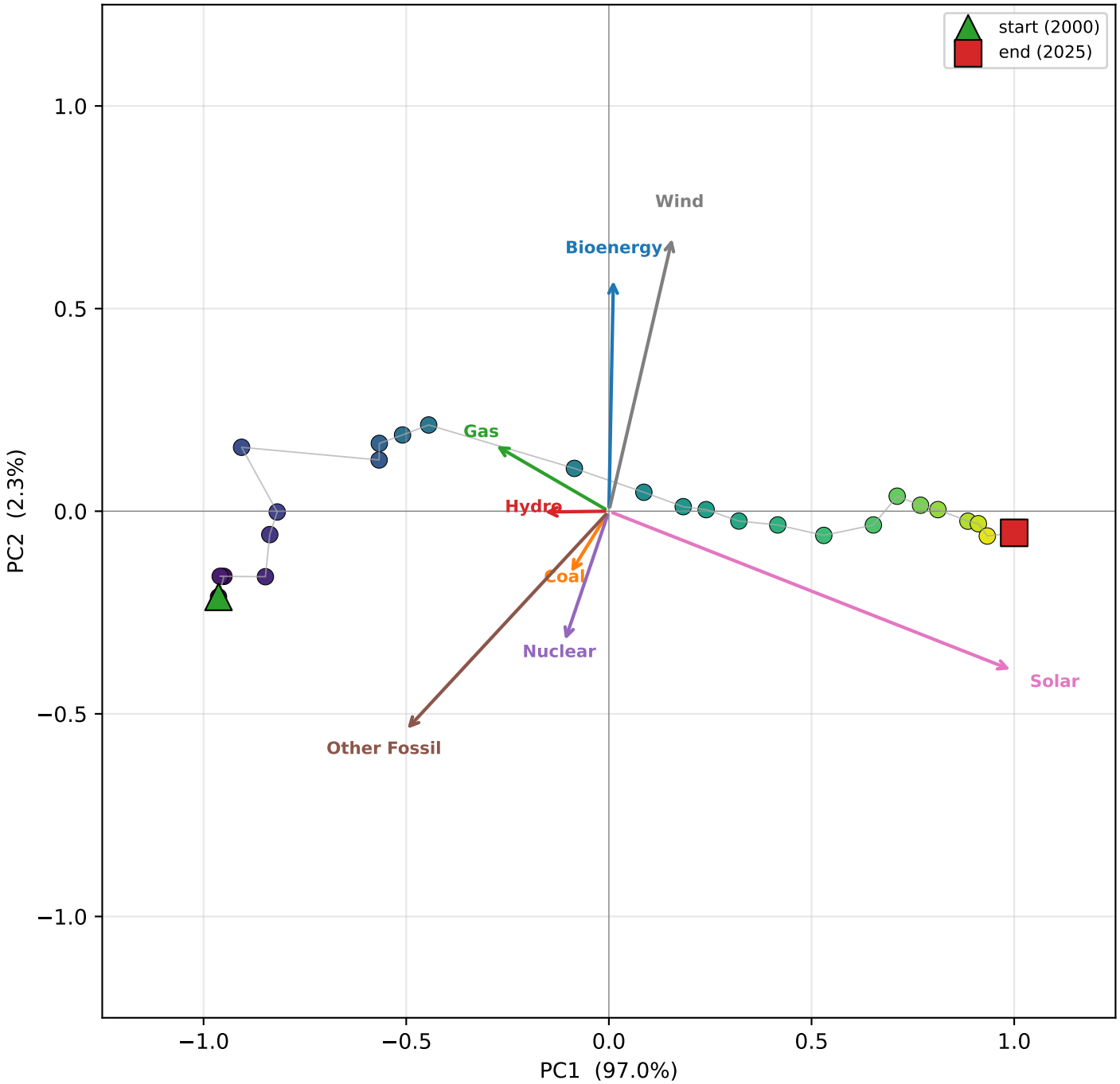

```
PC1: 96.98%   cum 96.98%
PC2:  2.26%   cum 99.24%
PC3:  0.34%   cum 99.58%
PC4:  0.22%   cum 99.80%
PC5:  0.12%   cum 99.92%
PC6:  0.05%   cum 99.97%
... 2 more PCs
```

Scree plot — eigenvalues of the centred CLR covariance matrix. | Computed via SVD of the centred CLR matrix (no reweighting). |

STAGE 2 | Plate 23 | CLR Biplot (CoDa biplot)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

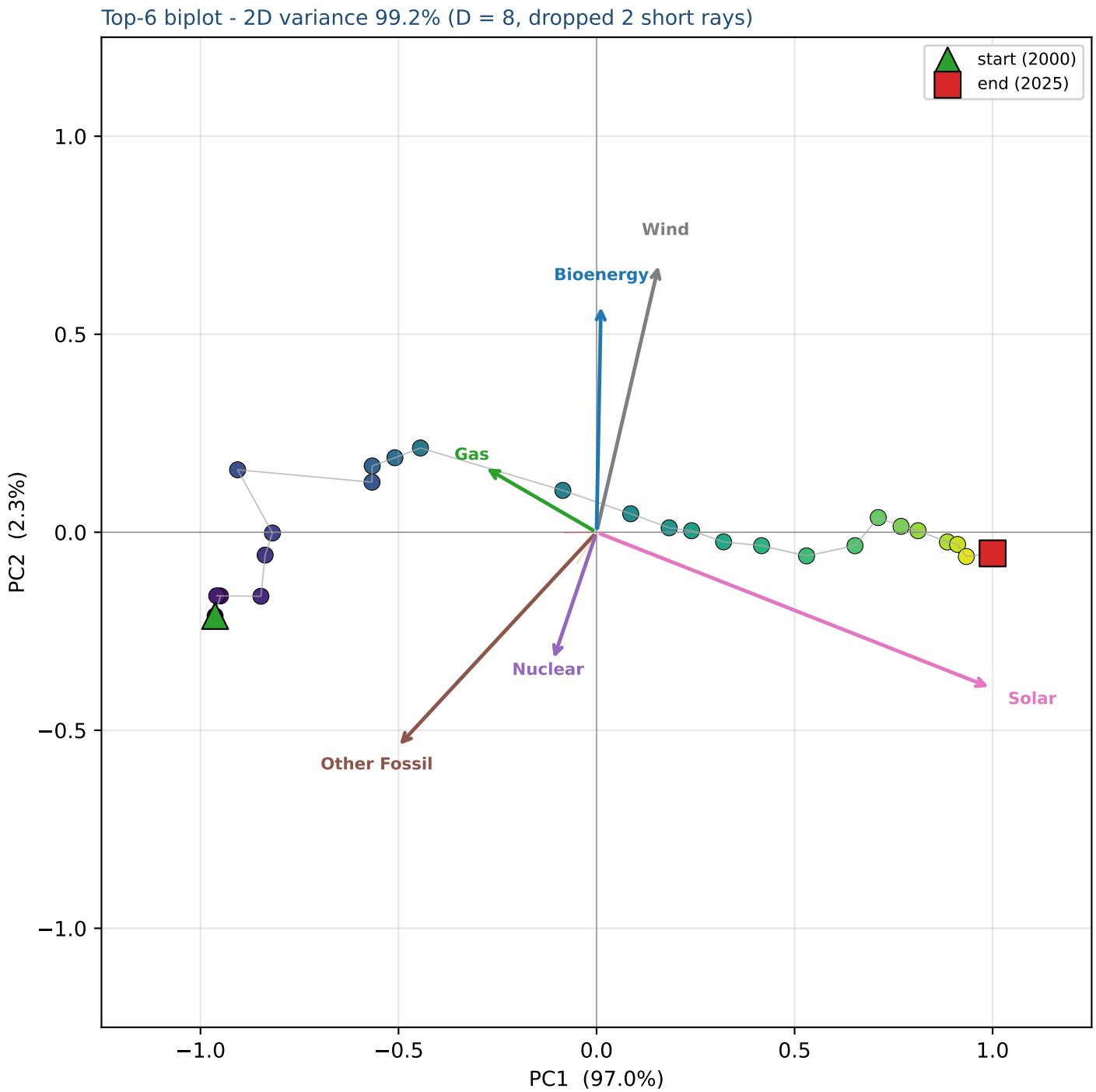
CoDa biplot — covariance form, 2D variance explained 99.2%




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 97.0% of variance.
PC2 captures 2.3% of variance.
Total 2D plane:  99.2%
```

STAGE 2 | Plate 24 | CLR Biplot - top-6 carriers by loading magnitude

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

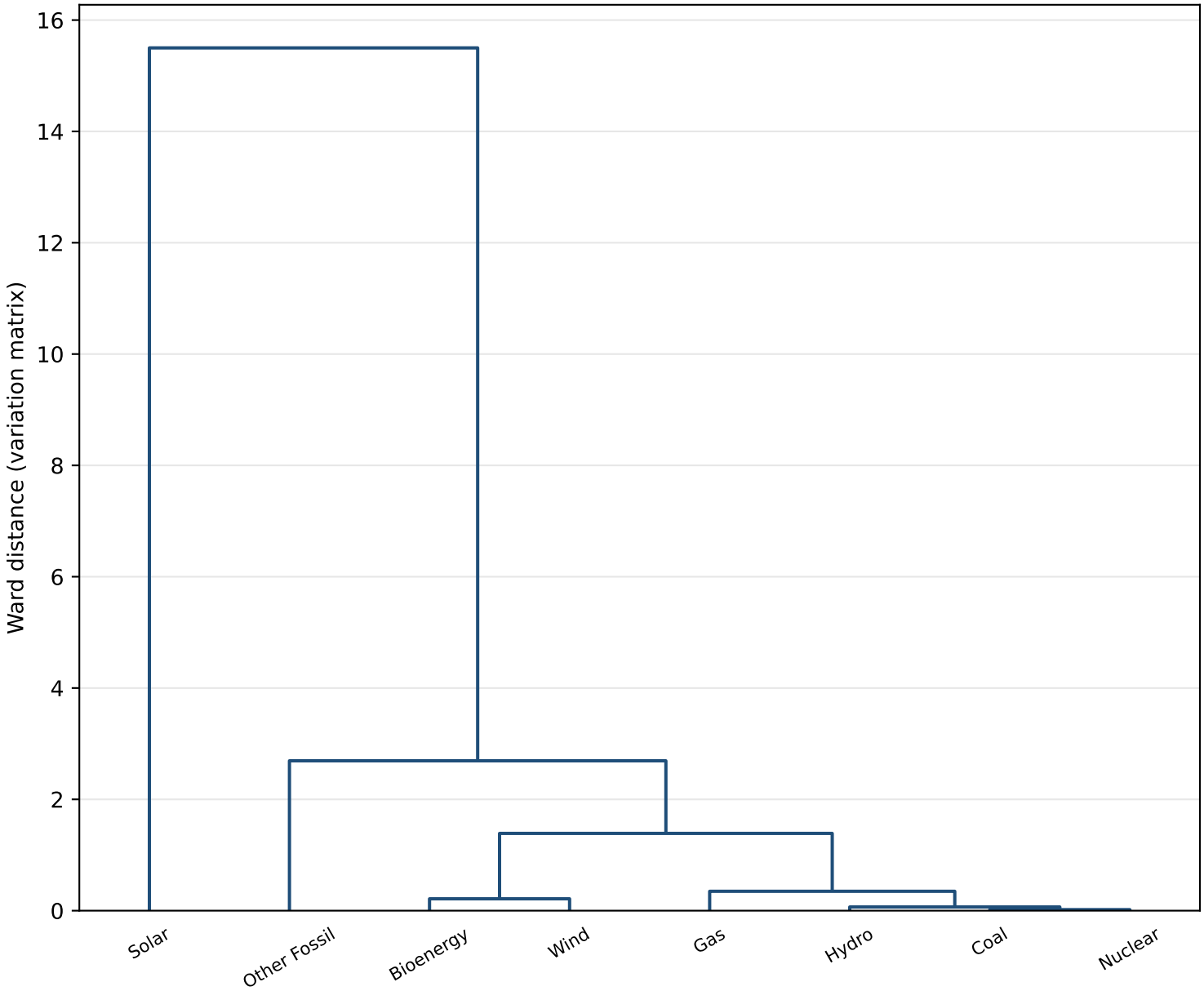



```
Loading magnitudes sqrt(G1^2 + G2^2):

  + Solar          0.9045
  + Other Fossil   0.6246
  + Wind           0.5894
  + Bioenergy      0.4870
  + Nuclear        0.2909
  + Gas            0.2787

Dropped (faint stubs only):
    Coal           0.1578
    Hydro          0.1419
```

STAGE 2 | Plate 25 | Balance-Dendrogram (Ward on variation matrix)

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 8 carriers → D-1 = 7 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

STAGE 2 | Plate 26 | Sequential Binary Partition (SBP) Matrix

ember_ind | Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, − = denominator, · = not in this balance

	b1	b2	b3	b4	b5	b6	b7
energy	·	·	+	·	+	−	−
Coal	+	−	·	−	−	−	−
Gas	·	·	·	+	−	−	−
Hydro	·	+	·	−	−	−	−
uclear	−	−	·	−	−	−	−
Fossil	·	·	·	·	·	+	−
Solar	·	·	·	·	·	·	+
Wind	·	·	−	·	+	−	−

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) where A_k = numerator set (+ rows), B_k = denominator set (- rows). D = 8; D-1 = 7 bal

```
T = 26,  D = 8
carriers: Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Solar, Wind

Navigation (System Course Plot):
  net_distance     = 9.4603 HLR
  path_length      = 14.6188 HLR
  course_directness= 0.6471

ω (deg)        : min 0.00  mean 4.87  max 11.11
Hs             : min 0.4746  mean 0.5161  max 0.5660
Aitchison norm : min 3.9973  mean 5.8979  max 8.7110
κ trace        : min 38737.71  max 4849064313.41
condition #    : min 1.14e+02  max 5.05e+04

Bearings: range [-179.55, 179.78]°,  std 96.53°
Lock events    : 0
Pairs          : 28
|r| > 0.7 pairs: 21

Helmsman frequency (top 5):
  Solar           13  (50.0%)
  Gas              4  (15.4%)
  Other Fossil     3  (11.5%)
  Wind             2  (7.7%)
  (none)           1  (3.8%)

Variation matrix trace : 0.0000
```